

Lab Assignment-10

Topic: Clustering Techniques (K-Means and DBSCAN)

Question 1: K-Means Implementation and Analysis

Implement the **K-Means clustering algorithm from scratch** (without using built-in clustering functions).

Tasks:

1. Generate a 2D synthetic dataset (e.g., using `make_blobs`).
2. Run the algorithm for different values of K (e.g., $K = 2, 3, 4, 5$).
3. Plot the clusters along with their corresponding centroids.
4. Compute the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

5. Use the Elbow Method to determine the optimal number of clusters.

Expected Outcome: Understanding of centroid initialization, convergence, and cluster compactness.

Question 2: K-Means vs DBSCAN (Comparative Study)

Use the non-spherical clusters dataset `make_moon`

Tasks:

1. Apply K-Means clustering and visualize the results.
2. Apply DBSCAN clustering with suitable values of ϵ and `min_samples`.
3. Compare the clustering outputs visually and quantitatively.
4. Discuss:
 - Limitations of K-Means for non-linear cluster shapes

- How DBSCAN handles noise and irregular clusters

Expected Outcome: Understanding the strengths and weaknesses of different clustering techniques.

Question 3: DBSCAN Parameter Sensitivity Analysis

Use a standard library (e.g., `sklearn`) to implement DBSCAN.

Tasks:

1. Apply DBSCAN on a dataset of your choice.
2. Experiment with different values of: ϵ (neighborhood radius) and `min_samples`
3. Plot the clustering results for each parameter setting.
4. Identify the Core Points, Border Points, and Noise Points
5. Analyze how parameter changes affect cluster formation.

Expected Outcome: Insight into density-based clustering behavior and parameter tuning.

Question 4: Real-world Dataset Clustering

Select the Customer Segmentation dataset from Kaggle.

Tasks:

1. Perform data preprocessing (handling missing values, normalization, etc.).
2. Apply both K-Means clustering and DBSCAN
3. Evaluate clustering quality using metrics such as Silhouette Score and visualize the clusters (2D/3D as applicable).
4. Provide a comparative analysis:
 - Which algorithm performs better and why?
 - Practical interpretation of clusters

Expected Outcome: Ability to apply clustering techniques to real-world datasets and interpret results.